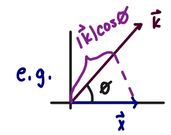
making waves

multi-dimensional waves $\left(e^{i(\vec{k}\vec{x}-\omega t)}\right)$

$\vec{k}\cdot\vec{x} = k_x x + k_y y + k_z z = |k||x|\cos\varnothing$

e.g.

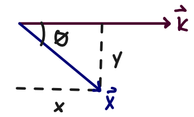


take $\vec{k}$ along $\vec{x}$

$\vec{k} = (1,0,0)$

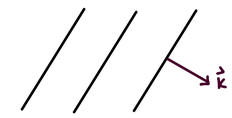
$\vec{x} = (x,y,z)$

$kx = x$



$k_x = kx$

$|k_x|\cos\varnothing$



$(\vec{k}\cdot\vec{x} - \omega t)$ keep only part of $\vec{x}$ along $\vec{k}$

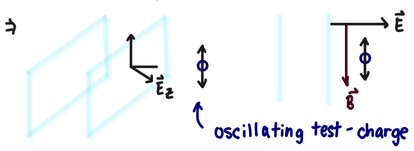
$\vec{k} = \boxed{\frac{2\pi}{\lambda}}\,\hat{k}$ ← wavelength

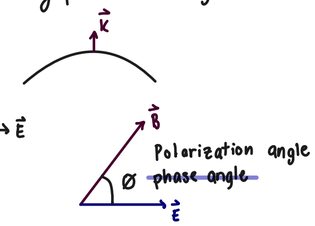
E & M waves

"vector field"

$\frac{d^2\vec{E}}{dt^2} = c^2\nabla\vec{E}$ $\left[\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right]$ ⇒

now, we have polarization





$\vec{E} = (E_x, E_y, E_z)$ $\hat{q} = \alpha\hat{E}_1 + \beta\hat{E}_2$

Make it Quantum

Schrödinger Equation

$$i\hbar\frac{d\psi}{dt} = \underbrace{\frac{\hbar}{2m}\frac{d^2}{dx^2}}_{\text{kinetic energy operator}}\psi + \overbrace{U(x)}^{\text{potential energy operator}}\psi$$

sol^{n}: $\psi(x,t) = \psi_0 e^{i(kx+\omega t+\varphi)}$ ⇒

$\frac{d^2\psi}{dx^2} = \underbrace{(ik)^2}_{-k^2}\psi_0 e^{i(\dots)}$

$i\frac{d\psi}{dt} = i(-i\omega)\psi_0 e^{i(\dots)}$

$\omega = vk$ ← phase velocity

$: \frac{i\,d\psi}{dt} = \alpha\frac{d^2\psi}{dx^2}$

De Broglie Relations

$p = \hbar k$ ← planck's constant

$E = \hbar\omega$

$\hat{E} = i\hbar\frac{d}{dt}$

"operator": $\hat{p} = -i\hbar\frac{d}{dx}$

$* \underbrace{-\frac{\hbar}{2m}\frac{d^2}{dx^2}}_{\hat{k}} = \frac{(\hat{p})^2}{2m} = \underbrace{\hbar\omega}_{E}$

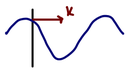
$\hat{p}\psi = -i\hbar(ik)\psi = (\hbar k)\psi$

Regular Wave Equation: $\underbrace{\frac{d^2}{dt^2}\psi_0 e^{i(kx+\omega t)}}_{(-i\omega)^2} = \underbrace{i^2\frac{d^2}{dx^2}\psi_0 e^{i(kx-\omega t)}}_{(ik)^2}$

Schrödinger: $i\hbar(-i\omega)e^{i(\dots)} = -\left(\frac{\hbar}{2m}\right)(ik)^2$

$v = \frac{\omega}{k}$ (dispersion relation)

$\left[\underbrace{k\omega}_{\omega}\psi = -\frac{\hbar^2}{2m}k^2\psi\right]$ ($\psi = e^{i(kx-\omega t)}$) ⇒ $\omega = \left(\frac{\hbar k}{2m}\right)k$



$E = -\hbar\omega$

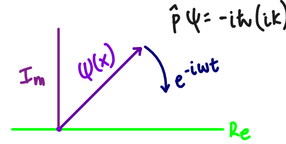
$p = \hbar k$

$\hbar k = p = mv$ $v = \frac{\hbar}{m\lambda}$

$\frac{mv}{2m} = \frac{v}{2}$

$\frac{p^2}{2m} = \frac{1}{2}mv^2$

phasor diagram:



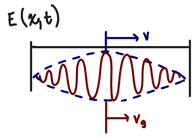
$\psi(x,t)$

$\psi(x)e^{-i\omega t}$

"unitary behind"

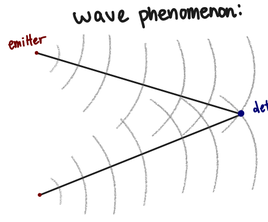
$\cos(kx-\omega t) = \frac{e^{i(kx-\omega t)} + e^{-i(kx-\omega t)}}{2}$

$\omega = (v(t))k$



$v_g = \frac{d\omega}{dk}$ Schrö: $\frac{d\omega}{dk} = \frac{2\hbar k}{2m} = \frac{p}{m} = \frac{mv}{m} = v$

wave phenomenon:



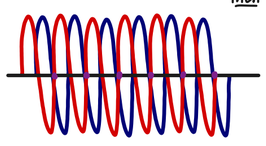
$e^{i(kx-\omega t+\psi)}$ ('phase')

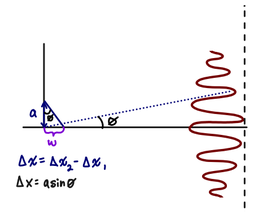
@ detector: sum

$e^{i(k\Delta x-\omega t)} + e^{i(k\Delta x-\omega t)}$

(Re part): $\cos(\underbrace{k\Delta x_1 - \omega_1}_{\varnothing_1}) + \cos(\underbrace{k\Delta x_2 - \omega_2}_{\varnothing_2})$

Intensity $\propto A^2$ - rule: first add ψ then square result





$\Delta x = \Delta x_2 - \Delta x_1$

$\Delta x = a\sin\varnothing$

Condition for constructive interference

↳ $k\Delta x = 2\pi N$

very far away -- "Fraunhofer" limit

$k\Delta x = 2\pi N$

$\left(\frac{2\pi}{\lambda}\right)a\sin\varnothing = 3\pi N$

$\sin\varnothing = \left|\left(\frac{\lambda}{a}\right)N\right|$, $N = 0, 1, \dots$

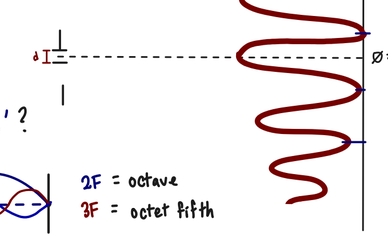
conditions for destructive interference

$k\frac{d}{2}\sin\varnothing = \pi N$

↳ $\frac{2\pi}{\lambda}\frac{a}{2}\sin\varnothing = \pi N$

$\varnothing = 0$: $\frac{\sin(0)}{0} \to \sin\varnothing = \left(\frac{\lambda}{d}\right)N$

$\lim_{x\to\infty}\frac{\sin x}{x} = 1$



$e^{i(kx-\omega t+\phi)}$ ('phase')

"boundary conditions"

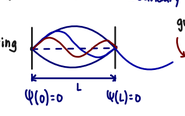
guess solution: $e^{i(kx-\omega t)} - e^{i(-kx-\omega t)}$

"standing wave" $e^{-i\omega t}(e^{ikx} - e^{-ikx}) = e^{-i\omega t}\,2i\sin kx$

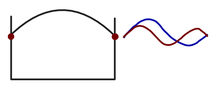
$e^{ikx} = e^{-ikx}e^{-i\omega t}$

$\cos kx + i\sin kx - [\cos kx - i\sin kx] = 2i\sin kx$

guitar string



"Particle in a box"



Does it satisfy 'Boundary conditions'?

$\sin kx = 0$ @ $x = 0$

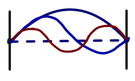
$\sin KL = 0$

$KL = \pi N$

$\frac{2\pi}{\lambda} = \frac{\pi N}{L}$

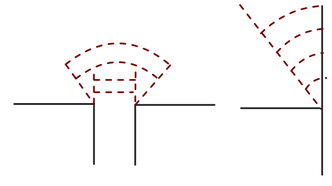
→ $\lambda = \frac{2L}{N}$ ← integer

'quantized' λ's

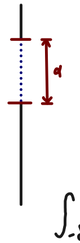


2F = octave

3F = octet fifth



diffraction:



Huygen's "wavelets"

sum → integral

complex relation

$e^{iy} = \cos y + i\sin y$

$$\int_{-\frac{d}{2}}^{\frac{d}{2}} dx\, e^{\overbrace{ikx\sin\varnothing}^{k\Delta x}} = \frac{d}{2}\,\frac{e^{ik\frac{d}{2}\sin\varnothing} - e^{ik\frac{d}{2}\sin\varnothing}}{ik\frac{d}{2}\sin\varnothing}$$